THEORY QUESTIONS AND ANSWERS

Question 1 (SSCE 1996 Qu 3)

a) What is a *skeleton*?

b) Name the **main** types of skeleton found in animals and for **each** type, give **one** example of an animal in which it is found.

c) What are the functions of the mammalian skeleton?

d) Using a suitable example, discuss the role of skeletal joint in movement in mammals.

Solution

a) ***Skeleton*** is the hard part of an animal which forms the framework of the body.

b)

Type of skeleton	*Example*
Endoskeleton	Mammals like goat, man, bat, whale
Exoskeleton	Arthropods like ant, crab, spider, millipede

c) ***Functions of the mammalian skeleton***

- Maintains the shape and form the rigid framework of the organism.
- Red blood cells are manufactured in the bone marrow.
- Provides surfaces for attachment of muscles.
- Provides a series of levers for movement of the body and its parts.
- Protects delicate internal organs.
- Aids the breathing mechanism.
- Supports internal organs as well as the total body weight of the organism.

d) ***Role of skeletal joint in movement in mammals***

- The role of skeletal joint in movement is to permit free movement without injury to the bones.
- Where two or more bones meet there is a joint, with the bones held loosely together by muscles and ligaments.
- Articulating surfaces of the bones are covered with cartilage to prevent wearing of the bones.
- There is the synovial membrane in between the bones which secretes the synovial fluid that acts as a lubricant.
- Before movements occur in the forearm, there must be two muscles that are antagonistic and must have point of origin to an immovable bone and point of insertion to a movable bone.
- The forelimb has its biceps originating from the scapula and point of insertion on the radius.
- The triceps has its origin at the scapula and the head of the humerus and its point of insertion on the ulna.
- The arm is straightened when the triceps contracts and the biceps relaxes.
- When the biceps contracts and triceps relaxes the lower arm is raised.

Question 2 (SSCE 2000 Qu 1)

a) i) Describe the girdles found in the mammalian skeleton.

ii) Name the type of joint formed by the girdles with limbs.

iii) Describe, without diagrams, the type of joint in **one** of the girdles you named in (ii) above.

b) How does **one** of the girdles in the mammalian skeleton bring about movements?

Solution

a) i) The two types of girdles are the *pectoral* and *pelvic* girdles.

Description of the pectoral girdle

- Each half consists of a scapula blade, coracoid and clavicle bone.
- The scapula is triangular in shape fused with the coracoid.
- The scapula has a cavity called glenoid cavity, into which fits the head of the humerus.
- One end of the clavicle articulates with the scapula, while the other end is joined by ligaments to the sternum.

Description of the pelvic girdle

- It consists of a pair of innominate bones such as ilium, ischium and pubis.
- The pelvic girdle is fused dorsally with the sacrum of the vertebral column and meets in front to form the pubic symphysis.
- On the side of each innominate bone is a large cavity called the acetabulum, into which the head of the femur fits.
- The pubis and ischium enclose a large hole called obturator foramen.

ii) The two girdles form a ***ball and socket joint*** with the fore and hind limbs.

iii) ***Ball and socket joint in pectoral girdle***

- The two articulating bones are humerus and pectoral girdle.
- The surfaces of the articulating bones are covered with articular cartilage, which prevent wearing off of the bones.
- Separating the bones is a synovial membrane which encloses a synovial cavity that secretes synovial fluid for lubricating the joint.
- The capsular ligament joins bone to bone.
- The head of the humerus fits into the socket of the glenoid cavity.

OR

Ball and socket joint in pelvic girdle

- The two articulating bones are femur and pelvic girdle.
- The surfaces of the articulating bones are covered with articular cartilage, which prevent wearing off of the bones.
- Separating the bones is a synovial membrane which encloses a synovial cavity that secretes synovial fluid for lubricating the joint.
- The capsular ligament joins bone to bone.
- The head of the femur fits into the socket of the acetabulum.

b) ***How the pectoral girdle bring about movement***

- The biceps and triceps are antagonistic muscles.
- Both muscles having their points of origin on the scapula and their points of insertion on the ulna and radius.
- When the biceps contract, the lower arm is raised and when the triceps contract, the lower arm is straightened.

OR

How the pelvic girdle bring about movement

- The two muscles involved are the extensor and flexor muscles.
- Both have their origins on the pelvic girdle and insertion on the tibia and fibula.
- Contraction of the extensor muscle straightens the knee joint and the leg moves forward.
- When the flexor muscle contract, the knee joint is bent and the leg moves backwards.

Question 3 (WASSCE Nov 2010 Qu 1)

a) i) Name the skeletal tissues found in *mammals*.

b) i) Describe the bone of the lower arm of a human.

ii) Outline the functions of the bones named in (*b*) (i).

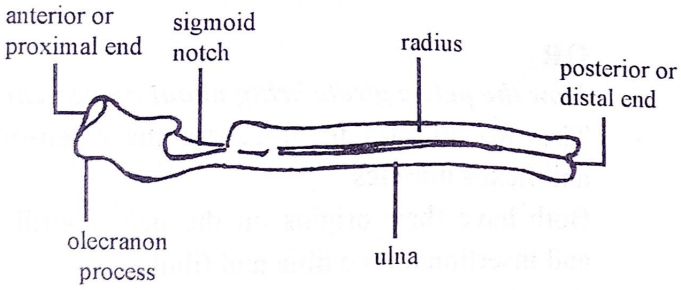
Solution

a) i) ***Skeletal tissues found in mammals***
- Bone
- Cartilage

b) i) ***Description of the bone of the lower arm of a human***
- It consists of radius and ulna. The ulna is longer and bigger than the radius with a depression called the sigmoid notch near its anterior or proximal end.
- Beyond the sigmoid notch is the olecranon process which forms below the elbow.
- Lying beside the ulna on the inner surface is a narrower bone slightly shorter than the ulna called the radius.

OR

Drawing of a lower arm bone of a human



Note: A well labelled diagram of a lower arm bone of a human is also accepted as an alternative answer.

ii) ***Functions of bones in the lower arm bone of a human***
- The head of the trochlea articulates with sigmoid notch to form a hinge joint which allows the arm to move in only one plane.
- The olecranon process provides surface for attachment of the triceps muscles.
- The radius provides a base for the attachment of the biceps muscles.

Question 4 (WASSCE Nov 2016 Qu 5e)

Describe the structure of the *cartilage* found in mammals.

Solution

Structure of cartilage in mammals
- It is a tough, flexible (fibrous/elastic/gelatinous) type of connective tissue with great tensile strength.
- The cartilage is made up of living cells within an extracellular matix of chondrin consisting chiefly of carbohydrate and protein fibres.
- The matrix is secreted by specialized cells called chondroblasts found in the cartilage. These cells are encycled by lacuna (a type of space).
- Within the matrix may also be found collagen fibres.

Question 5 (WASSCE June 2017 Qu 5d)

Describe **briefly** how skeletal muscles bring about movement of the limb of a mammal.

Solution

How skeletal muscles bring about movement of the limb of a mammal
- Muscles always work in pairs; these are biceps/flexor and triceps / extensor.
- These two sets of muscles work in opposition to each other (antagonistic in function) to bring about the bending and straightening of the limb.
- To straighten the arm, the biceps relaxes and the triceps contracts while in bending the arm, the biceps contracts and the triceps relaxes.
- Movement of these muscles is coordinated by the central nervous system.

Question 6 (WASSCE Nov 2018 Qu 5d)

i) Describe the structure of the cartilage found in mammals.

ii) Name **two** types of muscles found in mammals.

Solution

i) *Refer to* Question 4 (WASSCE Nov 2016 Qu 5e)

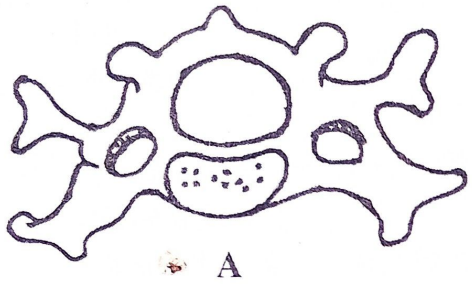
ii) ***Types of muscles found in mammals***

- Voluntary (skeletal / striated / striped) muscles
- Involuntary (smooth / unstriated / visceral) muscles
- Cardiac / heart muscle

PRACTICAL QUESTIONS AND ANSWERS

Question 1 (SSCE 1993 Qu 2a)

Specimen Provided:



A

Study carefully specimen **A** *and answer the questions that follow*:

Identify specimen **A** and describe **briefly** its features.

Solution

Identification of specimen **A**

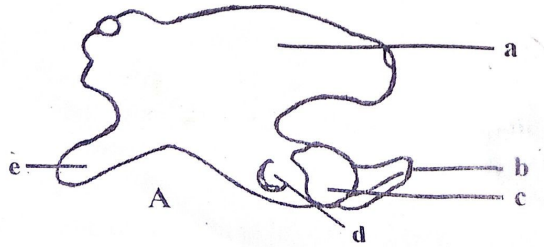
Cervical vertebra of a mammal

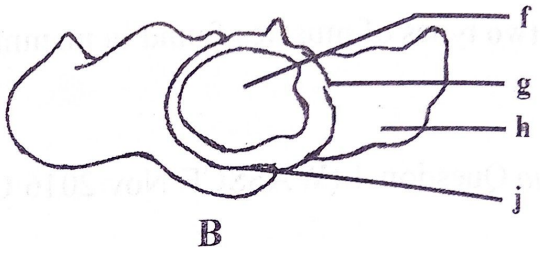
Features of specimen **A**

- It has centrum at the base.
- It has neural arch enclosing a neural canal.
- It has a reduced neural spine on top of the neural arch.
- Presence of pre- and post-zygapophysis for articulation with the next vertebra.
- It has two vertebrarterial canals on either side of the neural canal.

Question 2 (SSCE 1994 Qu 2a)

Study specimens **A** *and* **B** *carefully and answer the questions that follow.*





B

i) Give a heading to specimen **A**.

ii) Give a heading to specimen **B**.

iii) Name the features on the specimens labelled **a, b, c, d, e, f, g, h** and **j**.
State the function of **each** of the features labelled **b, g** and **h**.

Solution

i) ***Heading to specimen*** **A**
Lateral view of mammalian axis

ii) ***Heading to specimen*** **B**
Anterior view of mammalian atlas

iii) ***Names of labelled features***

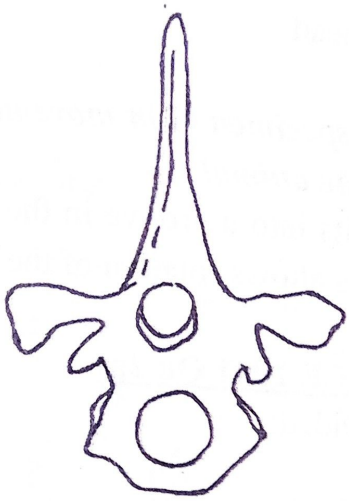
a – Neural spine
b – Odontoid process
c – Prezygapophysis
d – Vertebrarterial canal
e – Cervical rib
f – Neural canal
g – Prezygapophysis
h – Transverse process
j – Reduced centrum

Functions of labelled features

b – Fits into the lower part of the neural canal of the atlas
g – Articulates with the skull
h – For attachment of muscle

Question 3 (SSCE 1995 Qu 2a)

Specimen Provided:



A

Study carefully specimen **A** *and answer the questions that follow:*

i) Identify specimen **A**, giving your reasons.

ii) Make a labelled drawing, 4-6 cm long of the anterior view of specimen **A**.

Solution

i) ***Identification of specimen*** **A**

Mammalian thoracic vertebra

Reasons:

- Short transverse process
- Long neural spine
- Facet for articulation

ii) Drawing of anterior view of specimen A

Refer to notes on *Hints on Successful Biological Drawing* of specimens on page **v/vi**.

Question 4 (SSCE 1996 Qu 2a)

Specimen Provided:



A – Humerus of a mammal

Study carefully specimen **A** *and answer the questions that follow:*

i) Draw and fully label specimen **A**.

ii) State the functions of specimen **A**.

Solution

i) Drawing of a humerus of a mammal

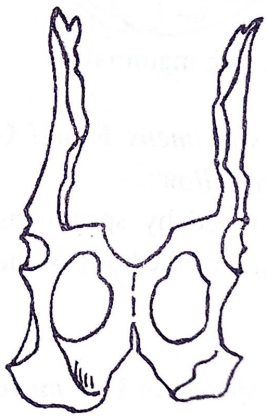
Refer to notes on *Hints on Successful Biological Drawing* of specimens on page **v/vi**.

ii) ***Functions of specimen*** **A**

- The trochlea articulates with the radius and ulna for movements of the lower arm.
- The head fits into the glenoid cavity of the scapula for the movement of the arm.
- Red blood cells are formed in its marrow.

Question 5 (SSCE 1997 Qu 2a)

Specimen Provided:



A – Pelvic girdle of a mammal

Study carefully specimen **A** *and answer the questions that follow:*

i) Draw and label fully specimen **A**.

ii) State the functions of specimen **A**.

Solution

i) Drawing of pelvic girdle of a mammal

Refer to notes on *Hints on Successful Biological Drawing* of specimens on page **v/vi**.

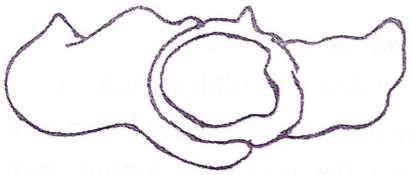
Functions of pelvic girdle of a mammal

Head of femur articulates with the acetabulum.

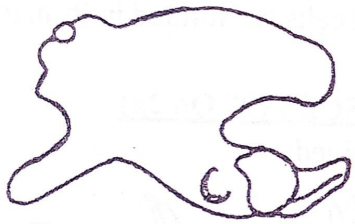
- It protects the female reproductive organs.
- Obturator foramen provides passage for blood vessels.
- Supports hind limbs.

Question 6 (SSCE 1999 Qu 2d)

Specimens Provided:



F – Atlas of a mammal



G – Axis of a mammal

Study carefully specimens **F** *and* **G** *and answer the questions that follow*:

State the role played by specimens **F** and **G** in movement of parts of the body of the animal.

Solution

Role played by specimen F in movement of parts of the body of the animal

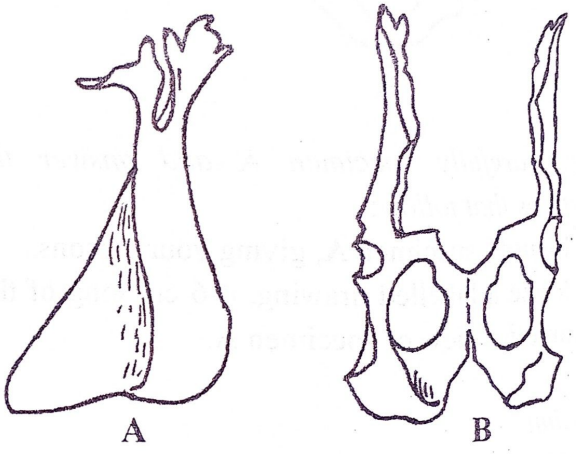
Concave surface of prezygapophysis or hollow anterior facet articulate with the skull, to allow nodding of the head.

Role played by specimen G in movement of parts of the body of the animal

Odontoid peg fits into a groove in the centrum of the atlas and this allows rotation of the head.

Question 7 (SSCE 2001 Qu 2a)

Specimens Provided:



A **B**

Study carefully specimens **A** *and* **B** *and answer the questions that follow*:

i) Identify specimens **A** and **B** without giving reasons.

ii) Copy and complete the table below, giving the details required for **each** of the specimens **A** and **B**.

Specimen	*Name of articulating surface*	*Description of articulating surface*	*Name of bone which articulates with specimen*	*Type of movement at the joint*
A				
B				

Solution

Identification of specimens

A – Scapula of a mammal

B – Pelvic girdle of a mammal

Specimen	*Name of articulating surfaces*	*Description of articulating surface*	*Name of bones which articulates with specimen*	*Type of movement at the joint*
A	- Glenoid cavity - Acromium process	- Shallow cavity - Flat	- Humerus - Clavicle	- Rotational movement - Slide movement
B	- Acetabulum - Inner upper surface of ilium	- Deep cavity - Flat	- Femur - Sacrum	- Rotational movement - Fixed joint so no movement

estion 8 (WASSCE June 2009 Qu 3e)

ecimen Provided:



L – Rabbit skull

udy carefully specimen **L** *provided and answer e following questions*:

) What type of joint is present in specimen **L**?

) State **three** functions of specimen **L**.

olution

) ***Type of joint present in rabbit skull***

Immovable joint consisting of fixed sutures

i) ***Functions of rabbit skull***

Protection of the brain, eyes, ear and nose.

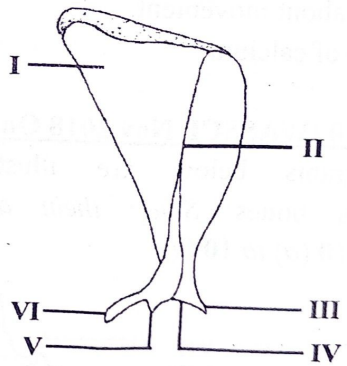
Gives shape to head.

Bears or supports the teeth.

- Articulates with the atlas to permit nodding.
- Provides attachment for lower jaw.

Question 9 (WASSCE Nov 2017 Qu 1)

The diagram below is an illustration of a mammalian bone. *Study it carefully and answer Questions (a) to (c).*



a) i) Identify the bone illustrated in the diagram.
ii) Name the parts labelled **I** to **VI**.

b) i) Which bone fits into the part labelled **IV**?
ii) Name the movable joint formed at **IV**.

c) State **two** functions of the bone illustrated in the diagram in mammals.

Solution

a) i) *Name of bone*
Scapula of a mammal or Shoulder blade of a mammal

ii) *Names of labelled parts*
I – Scapula blade
II – Spine
III – Coracoid process
IV – Glenoid cavity
V – Acromion
VI – Metacromion

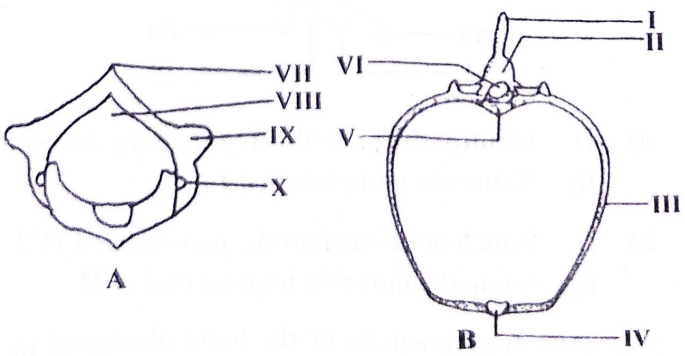
b) i) ***Bone which fits into the glenoid cavity***
Humerus

ii) ***Movable joint formed at glenoid cavity***
Ball and socket joint

c) ***Functions of the scapula in mammals***
- Gives support.
- Gives shape.
- Provides attachment of muscles.
- Brings about movement.
- Source of calcium.

Question 10 (WASSCE Nov 2018 Qu 3)

The diagrams below are illustrations of mammalian bones. *Study them and answer Questions* **10** (*a*) *to* **10** (*e*).



a) Identify:
i) the bone labelled **A**;
ii) the bones that make up the diagram labelled **B**.

b) i) Name the parts labelled **I** to **X**.
ii) Name **two** organs protected by the bones labelled **B**.
iii) In which part of the mammalian body are the bones labelled **A** and **B** located?

c) State **one function each** of the parts labelled **I**, **III**, **IV**, **VI**, **IX** and **X**.

d) i) State **six** differences between the bones labelled **A** and **B**.
ii) State **three** similarities between the bones labelled **A** and **B**.

e) Name the type of movement that occurs between the bone labelled **A** and:
(i) the bone above it;
(ii) the bone below it, in the skeletal system of humans.

Solution

a) i) ***Identification of bone labelled A***
- Atlas vertebra

Identification of bones that make up diagram B
- Thoracic vertebra
- Ribs
- Sternum (or breastbone)

b) i) *Names of labelled parts*
I – Neural spine
II – Neural arch
III – Rib
IV – Sternum
V – Centrum
VI – Neural canal
VII – Neural spine
VIII – Neural canal
IX – Transverse process
X – Vertebrarterial canal

ii) *Organs protected by the bones labelled* **B**
Heart and lungs

iii) Bone **A** (atlas vertebra) is located in the *neck* (or below the skull).

The bones labelled **B** are found in the *thorax* (or chest region).

c) ***Functions of labelled parts***

Labelled parts	*Functions*
I (Neural spine)	For attachment of ligament / muscles
III (Rib)	For respiratory movement; Protection of heart and lungs
IV (Sternum)	For attachment of ribs; For breathing
VI (Neural canal)	For passage of spinal cord
IX (Transverse process)	For attachment of muscles
X (Vertebrarterial canal)	Passage of blood vessels / nerves

d) i) ***Differences between bones* A *and* B**

A	**B**
Tuberculum absent	Tuberculum present
Rib absent	Rib present
Centrum absent	Centrum present
Sternum absent	Sternum present
Large neural canal	Small neural canal
Presence of vertebrarterial canal	Absence of vertebrarterial canal
Short / reduced neural spine	Long / prominent neural spine
Capitulum absent	Capitulum present
Transverse process prominent	Transverse process reduced

ii) ***Similarities between bones* A *and* B**
Both have:
- transverse process.
- neural spine.
- neural canal.
- neural arch.

e) i) Nodding (or up and down movement)

ii) Rotational movement